

Secondary Erythromelalgia: A Comprehensive Disease Characteristics Report

Summary

Secondary (acquired) erythromelalgia (EM) is a rare neurovascular acrosyndrome defined by the clinical triad of recurrent, paroxysmal **burning pain, erythema (redness), and increased local temperature (warmth) of the acral extremities**, characteristically provoked by heat, exercise, and limb dependency and relieved by cooling and elevation. Unlike **primary/inherited EM** (an autosomal-dominant *SCN9A*/Na_v1.7 sodium-channel gain-of-function neuropathy), secondary EM is not caused by a germline mutation. It is a **symptom of an underlying acquired disease** — most classically a **myeloproliferative neoplasm (MPN)**, especially essential thrombocythemia (ET) or polycythemia vera (PV), but also autoimmune/collagen-vascular disease (systemic lupus erythematosus), small-fiber peripheral neuropathy, diabetes mellitus, infections, and certain drugs (calcium-channel blockers, bromocriptine).

The best-characterized mechanism is the **MPN-associated (thrombocythemic) form**: somatic driver mutations (*JAK2* V617F, *CALR*, *MPL*) in hematopoietic stem cells produce clonal, hyperreactive platelets that drive **platelet cyclooxygenase (COX-1)/thromboxane-mediated arteriolar inflammation, fibromuscular intimal proliferation, and platelet-rich microthrombosis** of acral end-arterioles, producing localized ischemia experienced as burning pain, redness, and warmth. This form responds **dramatically and pathognomonically to low-dose aspirin** and remits with cytoreduction that normalizes the platelet count. A second, complementary line of evidence points to an **immune/complement-mediated microvascular injury and dysautonomia** mechanism (endothelial C5b-9 membrane-attack-complex deposition, type-I interferon upregulation, and diminished autonomic innervation of eccrine coils and arteries), linking EM to small-fiber neuropathy in non-MPN subtypes.

Clinically, secondary EM matters as a **sentinel of occult MPN**: symptoms can precede the hematologic diagnosis by a median of ~2.5 years, so all patients warrant serial blood counts and *JAK2* testing. Management is **cause-directed**: aspirin plus cytoreduction/phlebotomy for

MPN; drug withdrawal for drug-induced disease; neuropathic-pain agents and sodium-channel blockers for the small-fiber/autoimmune subtypes. Epidemiologically it is rare (secondary EM incidence $\approx 0.2/100,000/\text{year}$), female-predominant, and adult-onset, and is frequently unilateral/asymmetric — distinguishing it from the bilateral primary form.

Key Findings

1. Disease Information — Definition, Classification, and Identifiers

Erythromelalgia is a rare neurovascular acrosyndrome defined by a triad of recurrent burning pain, erythema, and increased temperature of the extremities, worsened by heat/dependency and relieved by cooling/elevation. It is divided into a **PRIMARY** form (inherited or idiopathic; autosomal-dominant *SCN9A*/Na_v1.7 gain-of-function neuropathy, typically early-onset, bilateral, and often aspirin-unresponsive) and a **SECONDARY** form (acquired; associated with an identifiable underlying disorder). Secondary EM associated with MPN classically responds dramatically to aspirin, a feature that both aids diagnosis and points to its mechanism.

"Secondary erythromelalgia is associated with myeloproliferative disorders, drugs (bromocriptine, calcium channel blockers), or clinical conditions such as rheumatic diseases or viral infection." — [PMID: 27639908](#)

"secondary, which is associated with myeloproliferative disorders-related thrombocythemia, polycythemia, collagen-vascular diseases, diabetes mellitus, peripheral neuropathy, autoimmune and infectious diseases, and use of certain medicaments" — [PMID: 15075045](#)

"secondary erythromelalgia associated with a myeloproliferative disease such as essential thrombocythemia often responds dramatically to aspirin therapy" — [PMID: 23862006](#)

Key identifiers (finding F011):

Resource	Identifier
MeSH	D004916 (Erythromelalgia)
ICD-10	I73.81 (Erythromelalgia)
ICD-11	EF41.2 / peripheral vascular category
Orphanet	ORPHA:90026 (Erythromelalgia)
MONDO	MONDO:0008574 (erythromelalgia)
OMIM	133020 — refers specifically to PRIMARY (SCN9A) erythromelalgia; secondary EM has no distinct OMIM entry because it is symptomatic of the underlying disease
HPO	HP:0012326 (Erythromelalgia)

Synonyms/alternative names: erythromelalgia, Mitchell disease, acromelalgia, "red limb" syndrome; secondary EM is also termed *acquired* or *symptomatic* erythromelalgia.

Information source type: Knowledge is derived predominantly from **aggregated disease-level resources and clinical case series/reviews** (Michiels series, Olmsted County population studies, single-center Swedish cohort), not from a single EHR-derived patient population. Secondary EM is defined nosologically as symptomatic of an underlying disease.

2. Etiology — Causal Factors, Risk Factors, Gene–Environment Interactions

Secondary EM is caused by a diverse set of underlying conditions (finding F009):

1. **Myeloproliferative neoplasms** — essential thrombocythemia and polycythemia vera (most common); also reported with chronic myelogenous leukemia (CML), primary/Philadelphia-positive myelofibrosis, and myelodysplastic syndrome when platelet counts are sufficiently high.
2. **Other hematologic disease** — thrombotic thrombocytopenic purpura (TTP).
3. **Autoimmune/collagen-vascular disease** — systemic lupus erythematosus (SLE), rheumatoid arthritis, vasculitis.
4. **Small-fiber peripheral neuropathy and diabetes mellitus.**
5. **Infectious/viral illnesses.**
6. **Drugs** — calcium-channel blockers (nifedipine, verapamil, diltiazem, felodipine), bromocriptine/dopamine agonists, and others.
7. **Rarely** — paraneoplastic/solid tumors and toxins.

"secondary, which is associated with myeloproliferative disorders-related thrombocythemia, polycythemia, collagen-vascular diseases, diabetes mellitus, peripheral neuropathy, autoimmune and infectious diseases, and use of certain medicaments... a single common pathogenetic mechanism - microvascular arteriovenous shunting" — [PMID: 15075045](#)

"We describe, for the first time, a patient in whom chronic myelogenous leukemia was associated with the development of erythromelalgia" — [PMID: 2643412](#)

"We describe a female patient in whom thrombotic thrombocytopenic purpura was associated with erythromelalgia. This has not been previously reported." — [PMID: 1613144](#)

Genetic risk factors: Secondary EM itself is not germline-determined. The relevant genetic lesions are the **somatic** driver mutations of the underlying MPN (see Section 4). *JAK2/MPL*-mutated ET carry higher arterial thrombosis risk, which is mechanistically upstream of the acral microthrombosis in EM.

Environmental risk factors: Older adult age and female sex (epidemiologic associations); drug exposure (calcium-channel blockers, bromocriptine) is the principal modifiable environmental cause. Environmental **provoking** (not causal) factors are heat, exercise, and limb dependency.

Protective factors: No established germline protective variants or dietary/lifestyle protective factors are described. The strongest "protective" intervention is pharmacologic — low-dose aspirin abolishes the thrombocythemic form.

Gene–environment interactions: The clearest interaction is between the somatic *JAK2/CALR/MPL* clone (genetic) and heat/dependency (environmental provocation): the clone establishes hyperreactive thrombocytosis, while environmental heat/dependency triggers the paroxysmal acral attacks superimposed on that substrate.

3. Phenotypes

The core phenotype (finding F007) is paroxysmal/episodic attacks of the triad:

Phenotype	HPO term	Notes / frequency
Severe burning pain of extremities	HP:0025267 (burning sensation); HP:0012531 (pain)	Cardinal symptom; provoked by heat/exercise/dependency
Erythema / red skin	HP:0000988 (skin rash); HP:0500064 (red skin)	Cardinal sign
Increased local skin temperature/warmth	— (warmth of acral skin)	Cardinal sign
Erythromelalgia (composite)	HP:0012326	Overall term
Acrocyanosis	—	With progression
Digital ischemia / necrosis / gangrene	—	Untreated MPN-associated disease

Distribution: Feet more than hands; often the ball of the foot and one or more toes or fingers. In secondary/MPN forms attacks are frequently **UNILATERAL or asymmetric** and **less intense** than in primary EM. **Provoking factors:** environmental heat, exercise, limb dependency. **Relieving factors:** cold exposure and elevation (patients immerse extremities in cold/ice water — a behavior that itself risks immersion/maceration injury).

"characterized by attacks of severe burning pain, erythema, and warmth of the extremities, primarily the feet and, to a lesser extent, the hands. The distress is provoked by environmental heat, exercise, and dependency; it is relieved by exposure to cold and elevation of the extremity" — [PMID: 2643412](#)

"characterized by red, congested distal extremities and painful burning sensations, usually confined to the ball of the foot and one or more toes or fingers. If left untreated, it may progress towards acrocyanosis and even peripheral gangrene" — [PMID: 11208311](#)

Associated features in MPN-associated cases: microvascular neurologic/ocular symptoms (headache, transient visual disturbance, paresthesias, TIA), pruritus, splenomegaly, and sometimes concurrent peripheral sensorimotor axonal neuropathy (documented in a *JAK2* V617F-positive PV patient, [PMID: 25674012](#)).

Age of onset: adult/late-onset (secondary form). **Severity:** variable (often milder than primary). **Progression:** episodic/fluctuating, potentially digit-threatening if untreated. **Quality-of-life impact:** severe during attacks — burning pain interferes with walking, sleep, footwear, work, and social function; compulsive cooling behaviors can cause secondary immersion skin injury.

4. Genetic / Molecular Information

Secondary EM is **not caused by a germline mutation**. Its most common underlying disease — MPN — is driven by **somatic, mutually exclusive driver mutations** in hematopoietic stem cells (finding F004):

Gene	HGNC/ OMIM	Frequency in ET	Frequency in PV	Functional consequence
<i>JAK2</i> (V617F)	OMIM 147796	~55%	>95%	Gain of function; constitutive JAK-STAT signaling
<i>CALR</i> (exon 9 indels)	OMIM 109091	~25%	rare	Gain of function; mutant CALR–MPL activation
<i>MPL</i>	OMIM 159530	~3%	rare	Gain of function; thrombopoietin-receptor activation
Triple-negative	—	~17%	—	—

Additional non-driver/clonal mutations occur: *TET2* (9–11%), *ASXL1* (7–20%), *DNMT3A* (~7%), *SF3B1* (~5%). *JAK2/MPL*-mutated ET is associated with higher arterial thrombosis risk — the pathophysiologic bridge to acral microthrombosis in EM.

"Approximately 80% of patients express myeloproliferative neoplasm driver mutations (*JAK2*, *CALR*, *MPL*), in a mutually exclusive manner" — [PMID: 38269572](#)

"*JAK2*, *CALR*, and *MPL* mutations are the mutually exclusive 'driver' mutations in ET with respective incidences of 55%, 25%, and 3%; approximately 17% are triple-negative" — [PMID: 27991718](#)

"the presence of *JAK2/MPL* mutations has been associated with higher risk of arterial thrombosis" — [PMID: 27991718](#)

Somatic vs. germline origin: entirely **somatic** for the MPN drivers; **allele frequency in population databases** is not the relevant metric (these are acquired clonal variants, not inherited polymorphisms), though clonal hematopoiesis of indeterminate potential (CHIP) with low variant allele frequency can complicate interpretation ([PMID: 38310350](#)). **Modifier genes / epigenetics / chromosomal abnormalities:** not specifically defined for the EM phenotype itself; abnormal karyotype and spliceosome mutations modify survival of the underlying MPN.

5. Environmental Information

Environmental factors / toxins: No established environmental toxin causes secondary EM directly. **Drugs** are the principal non-hematologic environmental cause — calcium-channel blockers (nifedipine, verapamil, diltiazem, felodipine) and bromocriptine/dopamine agonists (F009). **Lifestyle factors:** none causal beyond drug exposure; heat, exercise, and dependency are provocative triggers, not chronic risk factors. **Infectious agents:** viral/infectious illnesses are listed among secondary causes (F001/F009), but no single specified pathogen is established as causal; EM in this setting is best regarded as a reactive/para-infectious phenomenon.

6. Mechanism / Pathophysiology

Ordered causal chain — MPN-associated (thrombocythemic) secondary EM

1. A **somatic driver mutation** (*JAK2* V617F, *CALR*, or *MPL*) arises in a hematopoietic stem cell → **leads to** constitutive JAK2–STAT signaling and clonal myeloproliferation.
2. Clonal myeloproliferation **results in** thrombocytosis with **hyperreactive platelets** (essential thrombocythemia, or PV with high platelet counts).
3. Hyperreactive platelets, via **platelet cyclooxygenase-1 (COX-1)**, **lead to** increased thromboxane A₂ generation and platelet activation in acral end-arterioles.
4. Platelet COX/thromboxane activity **results in arteriolar inflammation and fibromuscular intimal proliferation** of dermal/subcutaneous arterioles.
5. This **leads to platelet-rich thrombotic occlusion** of the end-arterial microvasculature (skin biopsy: arteriolar inflammation, intimal proliferation, platelet thrombi; shortened platelet survival; elevated β -thromboglobulin, PF4, thrombomodulin; increased urinary thromboxane B₂).
6. Microvascular occlusion **results in** localized **acral ischemia** → **branch point:** experienced as **burning pain + erythema + warmth** (the EM triad); if severe/untreated → **acrocyanosis** → **digital necrosis/gangrene**.

7. **COX inhibition by aspirin/indomethacin** interrupts step 3 → complete, lasting symptom relief and normalization of platelet-activation markers (whereas sodium salicylate, dipyridamole, sulfinpyrazone, ticlopidine, and coumadin do NOT — establishing COX/thromboxane dependence). Cyto reduction that normalizes platelet count removes step 2 → remission; relapse recurs symptoms.

"Local platelet consumption in erythromelalgic areas became evident by the demonstration of arteriolar fibromuscular intimal proliferation and occlusions by platelet-rich thrombi in skin biopsies, by the findings of shortened platelet survival times, significant higher levels of platelet activation markers beta-thromboglobulin, thrombomodulin and increased urinary thromboxane B2 excretion" — [PMID: 12781799](#)

"is caused by platelet cyclo-oxygenase-mediated arteriolar inflammation, fibromuscular intimal proliferation without and with occlusive thrombosis by platelet-rich thrombi in the end-arterial microvasculature" — [PMID: 16510297](#)

"Skin punch biopsy samples taken from the affected areas showed typical arteriolar inflammation, fibromuscular intima proliferation, and thrombotic occlusions." — [PMID: 3977194](#)

Alternative / complementary causal chain — immune-mediated microvascular injury & dysautonomia (non-MPN subtypes)

1. An immune trigger (autoimmune disease such as SLE, or idiopathic dysimmunity) → **leads to** complement activation with **microvascular C5b-9 (membrane attack complex) deposition** on dermal endothelium.
2. Endothelial injury with **type-I interferon upregulation results in** microvascular dysfunction.
3. Concurrent **diminished autonomic innervation of eccrine coils and arteries** (CD56 evidence; mirroring small-fiber neuropathy) **leads to** dysregulated vasomotor control → maldistributed skin perfusion.
4. This **results in** the same clinical triad, but is **aspirin-unresponsive** and instead treated as neuropathic/autoimmune disease. (Mechanistic details are demonstrated in a small dermatopathology series and are partly *inferred*.)

"Biopsies showed superficial vascular ectasia in association with microvascular C5b-9 and variable upregulation of type I interferon expression in endothelial cells. CD56 stain revealed diminished autonomic innervation of the eccrine coil and arteries, mirroring similar autonomic denervation seen in small fiber neuropathy." — [PMID: 39846717](#)

"Histology commonly shows capillary proliferation, swelling of endothelial cells, perivascular edema, and chronic inflammation with sparse lymphocytic infiltrate." — [PMID: 23334520](#)

Unifying hypothesis: Ljubojević et al. proposed a **single common pathogenetic mechanism** across etiologies — microvascular **arteriovenous shunting** with maldistributed skin perfusion producing tissue hypoxia despite visible erythema and warmth ([PMID: 15075045](#)).

Ontology suggestions: - **GO biological processes:** platelet activation (GO:0030168), platelet aggregation (GO:0070527), blood coagulation (GO:0007596), prostaglandin biosynthesis / cyclooxygenase pathway (GO:0006693), inflammatory response (GO:0006954), complement activation (GO:0006956), type I interferon signaling (GO:0060337). - **CL cell types:** platelet (CL:0000233), blood vessel endothelial cell (CL:0000115), smooth muscle cell (CL:0000192). - **CHEBI:** thromboxane B2 (CHEBI:28728), acetylsalicylic acid/aspirin (CHEBI:15365), prostaglandin.

Mechanism schematic

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Somatic JAK2/CALR/MPL mutation (HSC)
  | constitutive JAK-STAT
  v
Clonal thrombocytosis (ET / PV)
  | hyperreactive platelets
  v
Platelet COX-1 -> thromboxane A2 ----(aspirin/indomethacin block here)
  |
  v
Arteriolar inflammation + fibromuscular intimal proliferation
  |
  v
Platelet-rich microthrombosis of acral end-arterioles
  |
  v
Acral ischemia --> BURNING PAIN + ERYTHEMA + WARMTH
  |
  +--> (untreated) acrocyanosis --> digital necrosis / gangrene

[Parallel non-MPN route]
Autoimmune/idiopathic trigger -> endothelial C5b-9 + type I IFN
+ autonomic denervation (eccrine/arteries) -> maldistributed
perfusion / AV shunting -> same triad (aspirin-unresponsive)

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7. Anatomical Structures Affected

Organ/system level (F010): primarily the **skin of acral extremities** and the **peripheral (microvascular) cardiovascular system**; the **peripheral/autonomic nervous system** is involved in neuropathic subtypes.

Localization (UBERON): feet (UBERON:0002387) > hands (UBERON:0002398); toes (UBERON:0009551) and fingers (UBERON:0002389); ball/sole of foot; distal/acral. Affected tissue: dermal/subcutaneous **arterioles** (UBERON:0001980) and capillaries.

Lateralization: secondary/MPN forms are frequently **unilateral or asymmetric** (versus bilateral in primary EM).

Cell level (CL): vascular endothelial cells (CL:0000115), platelets (CL:0000233) forming intraluminal thrombi; arteriolar smooth-muscle/intimal cells; eccrine sweat glands and their small autonomic (C-fiber) innervation in the dysautonomia/SFN subtype.

Subcellular level: platelet **cyclooxygenase (COX-1)** enzymatic activity; endothelial **complement (C5b-9)** and **type-I interferon** responses are the key molecular loci.

Secondary organ involvement/complications: acrocyanosis, digital ischemia, ulceration, gangrene; in MPN patients, concurrent cerebral/ocular/coronary microvascular territories (headache, visual disturbance, TIA).

"Skin punch biopsy samples taken from the affected areas showed typical arteriolar inflammation, fibromuscular intima proliferation, and thrombotic occlusions." — [PMID: 3977194](#)

"CD56 stain revealed diminished autonomic innervation of the eccrine coil and arteries" — [PMID: 39846717](#)

8. Temporal Development

Onset: adult/late-onset; often **insidious** in relation to the underlying MPN. In MPN-associated EM, symptoms **preceded** the diagnosis of the myeloproliferative disease by a **median of ~2.5 years** and can be the presenting manifestation of ET/PV (F008).

Progression: episodic/fluctuating course; untreated MPN-EM may progress to acrocyanosis and peripheral gangrene. **Duration:** chronic while the underlying disease persists; resolves when the cause is treated.

Remission patterns: treatment-induced remission is the rule — MPN-EM responds immediately to aspirin and remits with platelet-count normalization; drug-induced EM resolves on drug withdrawal. Symptoms **recur** with MPN relapse or aspirin withdrawal ([PMID: 22844295](#): EM developed within two weeks of aspirin withdrawal and improved within two weeks of restarting).

Critical period / window of opportunity: because idiopathic-appearing EM may be an early clonal MPN marker, **periodic blood counts** are recommended so that an abnormal hemoglobin/WBC/platelet count or immature cells prompts evaluation.

"symptoms of erythromelalgia preceded the onset of a myeloproliferative disease by a median of 2 1/2 years. Therefore, all patients with erythromelalgia should be monitored with periodic blood cell counts" — [PMID: 2643412](#)

"Her erythromelalgia immediately disappeared following interventional therapy along with aspirin." — [PMID: 15750823](#)

9. Inheritance and Population (Epidemiology)

Secondary EM is **rare, female-predominant, and adult-onset** (F002). The population-based Olmsted County study (Reed & Davis) reported an overall age/sex-adjusted EM incidence of **1.3/100,000/year** (95% CI 0.8–1.7), split into **primary EM 1.1** and **secondary EM 0.2/100,000/year**. Female incidence (2.0) exceeded male (0.6) per 100,000. The Swedish single-center study (Alhadad) found incidence 0.36/100,000/year, median age 49 (IQR 34–68), 70% women, and a mean diagnostic delay of 4.5 years; 3/27 developed intra-abdominal cancer.

"The incidence of primary and secondary erythromelalgia was 1.1 (0.7-1.5) and 0.2 (0.02-0.4) per 100,000 people per year, respectively." — [PMID: 18713229](#)

"we clinically identified 27 patients with EM. Median age was 49 [IQR (34 - 68)] years, 19 (70 %) were women" — [PMID: 22247059](#)

Inheritance pattern: secondary EM is **acquired/non-heritable**. The underlying MPN driver mutations are **somatic** (not inherited); there is no Mendelian inheritance, penetrance, anticipation, founder effect, consanguinity role, or carrier frequency applicable to secondary EM. (These concepts apply only to primary *SCN9A* EM.)

Population demographics: female predominance is confirmed across dermatologic epidemiology ([PMID: 27009931](#)). No specific ethnic/geographic clustering is established for secondary EM; distribution follows that of the underlying MPN/autoimmune diseases. **Sex ratio:** roughly 2–2.3:1 female:male. **Age distribution:** predominantly middle-aged and older adults.

10. Diagnostics

Diagnosis is clinical — the triad of burning pain, erythema, and warmth; heat/dependency provokes and cold/elevation relieves. Adjuncts include **infrared thermography, laser Doppler flowmetry, and cold-provocation testing**. The essential task is to **identify the secondary cause** (F005):

Test	Purpose
Complete blood count (CBC) with differential + peripheral smear	Detect thrombocytosis/erythrocytosis/leukocytosis (MPN)
<i>JAK2</i> V617F mutation testing (then <i>CALR</i> , <i>MPL</i> if negative)	Confirm MPN driver
Bone marrow biopsy	Confirm ET/PV/prefibrotic myelofibrosis
ANA	Screen for SLE/autoimmune cause
TSH	Thyroid disease
Small-fiber neuropathy testing (skin biopsy for intraepidermal nerve fiber density)	Neuropathic subtype
Screen for drugs / malignancy	Drug-induced and paraneoplastic causes
Skin punch biopsy	Arteriolar thrombosis (MPN), C5b-9/IFN, autonomic denervation

"For erythermalgia, a blood count and even a search for *JAK2* mutation are required. A thyroid-stimulating hormone assay, a test for antinuclear antibodies, and a search for small fiber neuropathy are also performed." — [PMID: 35835622](#)

Biomarkers / laboratory abnormalities: thrombocytosis (platelets $\geq 450 \times 10^9/L$; [PMID: 42101597](#)), erythrocytosis, elevated platelet-activation markers (β -thromboglobulin, PF4, thrombomodulin), increased urinary thromboxane B2.

Differential diagnosis (F011): Raynaud phenomenon (color change to white/blue, cold-induced — opposite triggers), complex regional pain syndrome, peripheral neuropathy, cellulitis, acrocyanosis, chilblains/pernio, Fabry disease acroparesthesias, gout, and **primary (SCN9A) EM**. The distinguishing feature of EM is **heat/dependency provocation and cold/elevation relief**.

"Raynaud's syndrome and Raynaud's mimickers, especially painful Raynaud's mimickers, can prove a diagnostic challenge" — [PMID: 38704280](#)

Genetic testing: for the **underlying MPN**, targeted *JAK2/CALR/MPL* testing (allele-specific PCR/NGS) is standard; WGS/WES are not routinely required. Note that low-*VAF JAK2* may reflect CHIP, and the dominant driver (e.g., *CALR*) determines phenotype (PMID: 38310350). No genetic testing is indicated for the EM phenotype itself in the secondary form.

11. Outcome / Prognosis

Overall EM mortality is low and not directly disease-attributable (Alhadad); however, the **underlying disease drives prognosis**. Untreated MPN-associated EM can progress to acrocyanosis and peripheral gangrene (digit-threatening). Prognosis is **favorable when the cause is treated**: aspirin gives immediate relief and cytoreduction produces remission; drug-induced EM resolves on withdrawal (F008).

The underlying MPN carries its own morbidity/mortality: PV median survival 14.1–27.6 years, with arterial thrombosis in 16% and venous thrombosis in 7% at/before diagnosis, ~12.7% progressing to myelofibrosis and ~6.8% to acute myeloid leukemia (PMID: 39556352); ET median survival ~18 years with leukemic transformation <1% at 10 years (PMID: 32974939). Thus EM's chief prognostic significance is as a **herald of occult MPN and its thrombotic/leukemic risk**.

Complications: digital ischemia, ulceration, gangrene, immersion/maceration skin injury from compulsive cooling; systemic thrombosis (arterial/venous) from the underlying MPN. **Prognostic factors:** platelet count/response to aspirin, presence and control of the underlying MPN, *JAK2/MPL* status (arterial thrombosis risk).

12. Treatment

Treatment is **cause-directed** (F005), with NCIT term suggestions:

Clinical scenario	Intervention	NCIT suggestion
MPN-associated EM	Low-dose aspirin (dramatic, near-pathognomonic relief)	NCIT:C287 (Aspirin)
MPN cytoreduction	Hydroxyurea (first-line)	NCIT:C512 (Hydroxyurea)
MPN cytoreduction (younger pts)	Interferon-α / pegylated interferon	NCIT:C20515 (Interferon Alpha)
PV	Therapeutic phlebotomy (goal hematocrit <45%)	NCIT:C15325 (Phlebotomy)
PV, HU-intolerant/resistant	Ruxolitinib (JAK inhibitor)	NCIT:C79809 (Ruxolitinib)
Drug-induced EM	Withdraw offending agent	—
Small-fiber/autoimmune EM	Neuropathic-pain agents (gabapentin, amitriptyline), sodium-channel blockers (mexiletine, lidocaine, carbamazepine) ; treat autoimmune disease	NCIT:C1367 (Gabapentin); NCIT:C614 (Lidocaine); NCIT:C1665 (Mexiletine)
Supportive (all)	Cooling, elevation, heat/dependency avoidance	—

"Aspirin is a useful treatment of erythromelalgia associated with myeloproliferative disorders. Treatment of primary erythromelalgia is difficult, individualized, with sodium channel blockers such as lidocaine, carbamazepine and mexiletine." — [PMID: 27639908](#)

The aspirin response is so characteristic that MPN cytoreductive guidelines (ET/PV updates, [PMID: 38269572](#), [PMID: 27561316](#)) routinely incorporate low-dose aspirin. In one illustrative case, reversal of thrombotic complications was achieved by aspirin and platelet reduction, "and not by coumadin" ([PMID: 16510297](#)). **Pharmacogenomics:** *JAK2/MPL* status guides thrombosis-risk stratification and cytoreduction decisions. **Personalized medicine:** treatment is genotype-/cause-guided rather than one-size-fits-all — the same clinical triad requires opposite drug classes (aspirin vs. sodium-channel blockers) depending on etiology.

13. Prevention

Because secondary EM is **acquired/symptomatic**, **no primary prevention exists** (F011). Prevention is framed at secondary and tertiary levels:

- **Secondary prevention:** early recognition of EM as a **sentinel of occult MPN**, with **serial CBC + JAK2 testing**, and treatment of thrombocytosis/erythrocytosis.
- **Tertiary prevention:** **aspirin ± cyto reduction/phlebotomy** to prevent digital necrosis and macrovascular thrombosis; **avoid provoking heat/dependency**; for drug-induced disease, **avoid/withdraw the culprit drug**.
- **Genetic counseling / carrier screening / immunization / public-health interventions:** **not applicable** (no germline/heritable or infectious-transmissible basis).

14. Other Species / Natural Disease

No naturally occurring secondary EM is described in companion animals or wildlife (OMIA); the disease is essentially **human** (F011). Orthologous genes for the underlying MPN drivers exist in model species (mouse *Jak2*, *Calr*, *Mpl*; NCBI Gene IDs available), enabling comparative study of the MPN substrate but not of the EM phenotype per se. There is **no zoonotic potential** and no cross-species transmission.

15. Model Organisms

No validated animal model of SECONDARY erythromelalgia exists (F011). Available models address only components of the disease:

Model	What it captures	Limitation for secondary EM
Transgenic/knock-in <i>SCN9A</i> /Na_v1.7 gain-of-function mice; patient iPSC-derived sensory neurons	Primary EM pain physiology	Models channelopathy, not acquired microvascular disease
<i>JAK2</i> V617F knock-in mice	Thrombocytosis/thrombosis of the underlying MPN	EM (acral burning/erythema) has not been specifically scored

Thus the field lacks a model that reproduces the platelet-COX-mediated acral microthrombosis or the immune/dysautonomic microvascular injury of human secondary EM — a significant translational gap.

Mechanistic Model / Interpretation

Secondary EM is best understood as a **final common acral microvascular phenotype reached by at least two distinct upstream routes**:

1. **The thrombocythemic (MPN) route** is the best-established and mechanistically the most complete: a somatic *JAK2/CALR/MPL* clone → hyperreactive thrombocytosis → platelet COX-1/thromboxane-driven arteriolar inflammation and platelet-rich microthrombosis → acral ischemia. The therapeutic "experiment of nature" — complete relief with COX inhibitors (aspirin/indomethacin) but not with anticoagulants (coumadin), antiplatelet agents acting by other mechanisms (ticlopidine, dipyridamole), or non-acetylated salicylate — pins the mechanism firmly on **platelet cyclooxygenase**. This is a distinct, aspirin-responsive **arterial thrombophilia**.
2. **The immune/dysautonomic route** (autoimmune, idiopathic, small-fiber-neuropathy-associated) involves **complement (C5b-9) and type-I interferon-mediated endothelial injury** plus **autonomic denervation** of eccrine glands and arterioles, producing maldistributed perfusion/arteriovenous shunting. This route is aspirin-**unresponsive** and treated as neuropathic/autoimmune disease.

Both converge on Ljubojević's proposed unifying lesion — **microvascular arteriovenous shunting with tissue hypoxia despite visible hyperemia** — reconciling the paradox of a red, warm, yet ischemic and painful extremity. The unifying clinical message is that **the triad is a syndrome, not a diagnosis**: identifying and treating the upstream cause is both diagnostic confirmation and definitive therapy.

Evidence Base

PMID	Title (abbrev.)	Role
12781799	Platelet-mediated microvascular inflammation and thrombosis in thrombocythemia	Core biochemical/histologic evidence for COX/thromboxane mechanism
3977194	EM caused by platelet-mediated arteriolar inflammation	Histopathology of arteriolar thrombosis
16510297	Reversal by aspirin, platelet reduction, not coumadin	Establishes COX/aspirin dependence
39846717	Cutaneous pathology of EM	Complement/IFN/dysautonomia mechanism (non-MPN)
23334520	Secondary EM, unusual histology	Typical histopathology
2643412	EM and myeloproliferative disorders	Phenotype, ~2.5-yr lead time, CML association, monitoring
11208311	EM — thrombotic complication in MPD	Localization and progression to gangrene
15075045	Erythromelalgia (review)	Etiologic spectrum + AV-shunting unifying mechanism
1613144	EM in TTP	TTP as secondary cause
38269572	ET 2024 update	MPN driver mutations (~80% JAK2/CALR/MPL)
27991718	PV/ET 2017 update	Driver frequencies; JAK2/MPL & arterial thrombosis
18713229	Incidence, Olmsted County	Population-based incidence (secondary 0.2/100k/yr)
22247059	Incidence, Sweden	Age/sex distribution, diagnostic delay
35835622	Paroxysmal vascular acrosyndromes	Diagnostic workup
27639908	EM diagnosis & therapy	Cause-directed treatment; drug causes
23862006	Secondary EM case report	Aspirin responsiveness
15750823	EM in ET / renovascular HTN	Rapid treatment response

PMID	Title (abbrev.)	Role
22844295	EM on aspirin withdrawal	Recurrence with aspirin withdrawal
38704280	Painful Raynaud's mimics	Differential diagnosis
25674012	Burning feet in PV	Concurrent axonal neuropathy in JAK2+ PV
39556352	PV review	Underlying MPN prognosis; EM frequency 5.3% in PV
38310350	Multiple driver mutations in ET	CHIP vs. dominant driver interpretation
42101597	Thrombocytosis review	Thrombocytosis definition/workup
32974939	PV/ET 2021 update	Survival, transformation rates
27561316	How I treat ET	Aspirin + cytoreduction strategy

Consistency: Multiple independent series (Michiels; Olmsted County; Swedish cohort; ET/PV guideline updates) converge on the same picture. The classic thrombocythemetic mechanism is strongly and repeatedly supported; the immune/dysautonomic mechanism rests on a smaller, more recent evidence base (single dermatopathology series) and should be regarded as an emerging but not yet fully validated complementary model.

Limitations and Knowledge Gaps

- **No dedicated secondary-EM animal model.** *JAK2* V617F mice reproduce the MPN substrate but EM has not been scored; there is no model of the immune/dysautonomic route.
- **Immune/complement mechanism is based on a single small series** (9 patients, [PMID: 39846717](#)) and requires independent replication with quantitative endpoints.
- **Epidemiologic estimates are geographically narrow** (Olmsted County, single Swedish center); global prevalence, ethnic distribution, and secondary-cause proportions are poorly characterized.

- **No prospective controlled trials** define optimal therapy for non-MPN subtypes; treatment recommendations are extrapolated from case series.
 - **Frequency data for individual phenotypic features** (e.g., proportion unilateral, proportion progressing to gangrene) in secondary EM specifically are sparse.
 - **The AV-shunting unifying hypothesis** remains conceptual and unproven at the tissue level for all subtypes.
 - **Molecular profiling (transcriptomics/proteomics/metabolomics) of EM-affected skin** is essentially absent beyond immunohistochemistry.
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Proposed Follow-up Experiments / Actions

1. **Prospective natural-history cohort** of idiopathic-appearing EM with serial CBC + *JAK2/CALR/MPL* testing to quantify the rate and lead time of occult MPN conversion and refine the monitoring interval.
 2. **Replicate the complement/interferon/dysautonomia findings** in an independent, larger biopsy cohort with quantitative C5b-9, type-I IFN signature scoring, and intraepidermal nerve-fiber-density measurement, stratified by MPN vs. non-MPN etiology.
 3. **Develop a secondary-EM model** by scoring acral thermal/nociceptive phenotypes and skin microvascular thrombosis in *JAK2* V617F knock-in mice $\pm$ aspirin, testing whether COX inhibition reverses acral pathology.
 4. **Aspirin-response as a diagnostic biomarker study:** formally test the sensitivity/specificity of a dramatic aspirin response for detecting underlying MPN.
 5. **Skin single-cell/spatial transcriptomics** of EM-affected vs. unaffected acral skin to map the endothelial, platelet, smooth-muscle, and autonomic-nerve contributions and validate the two-route model.
 6. **Registry-based epidemiology** across multiple regions to establish global incidence, secondary-cause proportions, and demographic distribution.
 7. **Trial of complement- or interferon-directed therapy** in refractory, aspirin-unresponsive, immune-associated EM, guided by biopsy biomarkers.
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Report compiled from 11 confirmed findings across 5 investigation iterations and 35 reviewed papers. Evidence sources are predominantly human clinical case series, population-based epidemiology, and disease-level reviews; mechanistic claims are supported by skin-biopsy histopathology, platelet biochemistry, and pharmacologic-response data.

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